CHAPTER D3

Integration

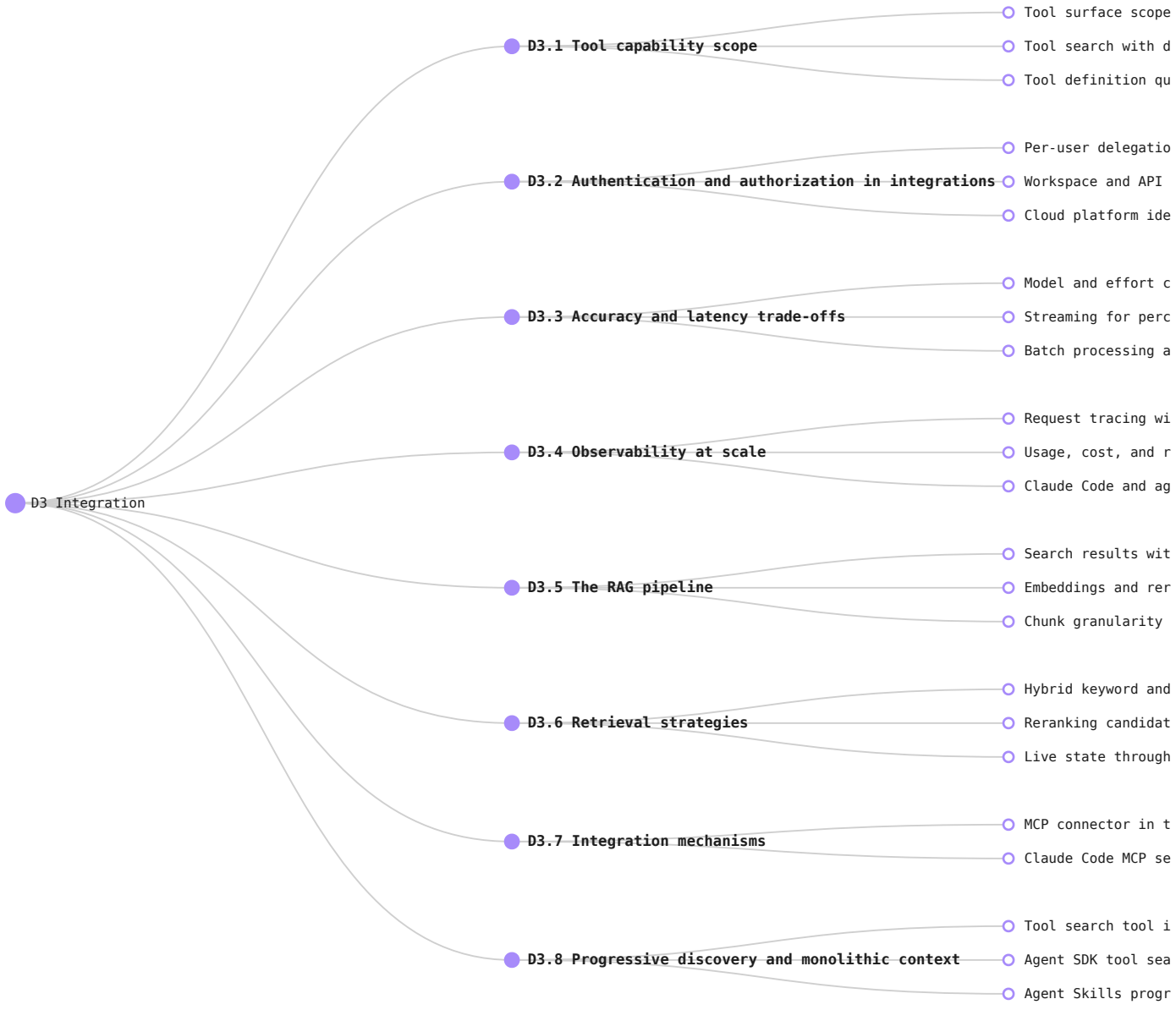
WEIGHT 19%. INFERENCE FROM SECONDARY SOURCES.

Scope tools, trace hops, ground answers, defer catalogs.

8 sections, 23 components, 24 recall cues.

CONCEPT MAP

Units branch to components. On a small screen, scroll sideways.



SECTIONS

D3.1 CHAPTER D3, INTEGRATION

Tool capability scope

Remove the tools the task does not need. Then monitor.

CONSTRAINT FAMILIES



TOOL SURFACE SCOPE

Each attached tool is a permission, a token cost, and a selection choice, so the set should hold only what the task calls.

Rule Remove or scope each tool the task does not need before you add any monitoring.

Trap Full audit logs on a broad tool set look like the safe fix. The permission stays attached, so the next wrong call still runs.

Exam tip If the scenario names an unneeded write capability, pick removal, a cap, or an approval gate.

TOOL SEARCH WITH DEFERRED TOOLS

Deferred tools stay out of the first prompt, and the API adds the found tools when the model searches for them.

Rule Defer each tool that few requests use, and keep the most used tools non-deferred.

Trap A longer system prompt that lists every tool looks like a fix. It keeps every schema in the first prompt.

Exam tip If the scenario shows 10 or more tools and high cost, pick tool search over a longer prompt.

TOOL DEFINITION QUALITY

Clear, few, and consistently named tool definitions give the model the fewest wrong picks.

Rule Give each job one tool name and one description, and merge tools that do the same job.

Trap A system prompt rule that says to pick the right search tool looks like a fix. Two tools with the same job still split the model choice.

Exam tip If the scenario shows two similar search tools, pick the option that merges them or renames them for distinct jobs.

▼ RECALL CUES (3)

1. **the agent picks the wrong tool after the tool list grows**
   Answer scope the tool set and merge similar names
   Rule Merge tools that do the same job and remove tools the task never calls.
   - Routing layer on top of the same tool set
   - Logging every call with no removal
2. **a wrong write fires through a tool the task rarely needs**
   Answer remove the write tool or cap its writes
   Rule Remove the write capability, or cap it and gate it with approval.
   - Keep all tools and log every call
   - Longer audit retention
3. **the tool schema cost per call rises with the library**
   Answer tool search with deferred tools
   Rule Defer tools that few requests use and keep frequent tools non-deferred.
   - Delete tools the task still needs
   - Longer system prompt listing every tool

▼ NUMBERS AND VERSIONS (12)

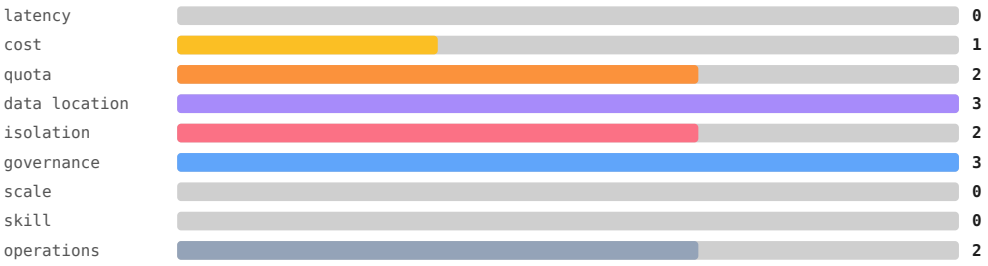
- A typical multiserver setup can consume about 55k tokens in tool definitions. source
- Tool search typically reduces definition tokens by over 85 percent. source
- Claude tool selection accuracy can degrade once the set exceeds about 30 to 50 tools. source
- Tool search is advisable with 10 or more tools. source
- Each search returns up to 5 found tools by default. source
- A request can define up to 10,000 tools with defer_loading set to true. source
- Regex patterns are limited to 200 characters, and BM25 queries to 500 characters. source
- Standard tool use fits when a request has fewer than 10 tools. source
- A deferred tool cannot carry cache_control, and the API returns a 400 error. source
- Tool descriptions should run at least 3 to 4 sentences for each tool. source
- Tool examples add about 20 to 50 tokens for simple examples. source
- A request can set strict on at most 20 tools. source

D3.2 CHAPTER D3, INTEGRATION

Authentication and authorization in integrations

Per-user credential when the audit must name the person.

CONSTRAINT FAMILIES



PER-USER DELEGATION TO MCP SERVERS

A remote MCP server authorizes the call from an OAuth token that the client sends for that server, so the token must name the user.

Rule Give each server a per-user credential when the audit must name the person who asked.

Trap A shared service key with a user name in the prompt looks like attribution. The system cannot verify a name that the user typed.

Exam tip If the scenario offers a session id or a prompt field as identity, reject it.

WORKSPACE AND API KEY BOUNDARY

A workspace groups keys, members, and limits, and a key scoped to one workspace reaches only that workspace.

Rule Scope the key to the one workspace that owns the data, and give members the least role.

Trap An unscoped key with a workspace header looks isolated. The key still reaches every workspace that its user or service account can access.

Exam tip A key scoped to one workspace cannot reach another workspace, even with a header that names it.

CLOUD PLATFORM IDENTITY

Each cloud platform checks its own identity and IAM policy before it runs a request, so the policy becomes the access control.

Rule Use the platform identity and scope the IAM policy to the workspace or deployment.

Trap A long-lived static key in code looks simpler. It bypasses the platform identity, so the IAM policy cannot scope or revoke each caller.

Exam tip If the scenario needs OAuth user consent on Claude Platform on AWS, reject that option because OAuth is not supported.

▼ RECALL CUES (3)

1. **the audit must name the person who asked for each action**
   Answer per-user credential checked at the server
   Rule Give each user a credential for each server, and never share one key for all users.
   - Shared service key with a name in the prompt
   - Department-level credential
2. **one key reaches several teams and tenants**
   Answer workspace scope on each key
   Rule Scope each key to one workspace and give members the least role.
   - Unscoped key with a workspace header
   - One shared workspace for all tenants
3. **the cloud platform must control who calls the model**
   Answer platform IAM policy on the workspace or deployment
   Rule Use the platform identity, and scope the IAM policy to the workspace or deployment.
   - Long-lived static key in code
   - OAuth consent on Claude Platform on AWS

▼ NUMBERS AND VERSIONS (4)

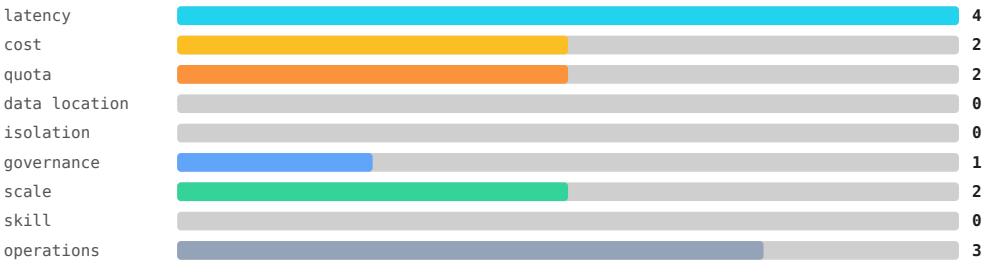
- Claude Code flags a remote server for sign-in on a 401 or 403 response. source
- An organization can hold up to 100 workspaces by default. source
- Bedrock bearer tokens have a 12-hour maximum and are the least preferred option. source
- The default Bedrock quota is 2 million input tokens per minute. source

D3.3 CHAPTER D3, INTEGRATION

Accuracy and latency trade-offs

Smallest model and effort that pass quality and latency.

CONSTRAINT FAMILIES



MODEL AND EFFORT CHOICE FOR LATENCY

Latency rises with model size, output length, and effort, so choose the smallest model and effort that pass the quality target.

Rule Pick the smallest model and effort level that passes the measured quality target and the latency target.

Trap A bigger model looks like the accuracy fix. It adds time per call and can break the latency target.

Exam tip When the scenario states a latency limit, reject an option that adds a slower step before you measure it.

STREAMING FOR PERCEIVED LATENCY

Streaming sends output as the model generates it, so users see the first tokens before the full response ends.

Rule Stream when a person reads the output as it arrives, and measure time to first token.

Trap Streaming looks like a generation speed-up. It shows output sooner, and the total generation time stays about the same. (Inference)

Exam tip If the scenario needs the full result before any step, streaming does not meet the target.

BATCH PROCESSING AND SERVICE TIERS FOR THROUGHPUT

Work that can wait trades response time for half the standard price, and the batch runs in the background.

Rule Use Batch when the caller accepts results within 24 hours and needs half the standard price.

Trap Batch looks like a speed fix for a large job. Results can take up to 24 hours, so it does not meet a seconds-level target.

Exam tip If the scenario needs an answer in seconds, reject Batch whatever the volume.

▼ RECALL CUES (3)

1. **the answer misses the accuracy target and breaks the latency target after several additions**
   Answer cut the stage with the weakest measured gain
   Rule Measure each stage and cut the one whose gain is smallest for its time cost.
   - Add a bigger model for more accuracy
   - Add a reasoning stage with no measured gain
2. **users wait too long for the first words of a reply**
   Answer streaming with time to first token measured
   Rule Stream the output and measure time to first token as its own metric.
   - Faster model without a measured need
   - Longer timeout in the client
3. **a large offline job needs results and the cost must fall**
   Answer Message Batches API for the offline set
   Rule Move work that can wait a day to Batch, and keep interactive calls on the standard path.
   - Parallel calls with standard pricing
   - Batch for an interactive chat

▼ NUMBERS AND VERSIONS (10)

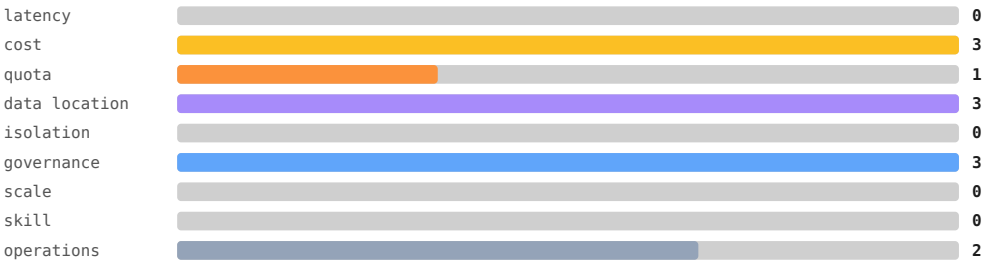
- Claude Haiku 5.5 defaults to medium effort. source
- Fast mode delivers up to 2.5 times higher output tokens per second. source
- Fast mode on Claude Opus 5.5 costs $8 per MTok input and $40 per MTok output. source
- After a 504 timeout error, the docs suggest the streaming Messages API for long requests. source
- The SDK validates that Messages requests without streaming are not expected to exceed a 10-minute timeout. source
- Batch usage is charged at 50 percent of standard API prices. source
- A batch holds up to 100,000 requests or 256 MB, whichever is reached first. source
- Most batches finish within 1 hour. source
- Batch results are available for 29 days after creation. source
- A batch expires if processing does not complete within 24 hours. source

D3.4 CHAPTER D3, INTEGRATION

Observability at scale

Trace every model and tool hop with one trace ID.

CONSTRAINT FAMILIES



REQUEST TRACING WITH OPENTELEMETRY

A trace records each step of one run with a shared trace ID, so you can rebuild the run from the spans.

Rule Trace each hop that calls a model or a tool, and tie the hops to one trace ID.

Trap More prompt logs or longer retention keeps the blind spot. The logs hold no tool or retrieval hop.

Exam tip If the scenario asks which tool call acted, pick the trace spans over the prompt text.

USAGE, COST, AND RATE LIMIT REPORTS

Usage and cost reports group token counts and dollars by workspace, key, and model, and response headers show live capacity.

Rule Use the usage and cost reports for spend, and response headers for live capacity.

Trap Totals of input_tokens from each response look complete. That field counts only tokens after the last cache breakpoint.

Exam tip A spend-cap 429 has no retry-after header and stays failed until access resumes.

CLAUDE CODE AND AGENT SESSION TELEMETRY

Claude Code exports its own metrics and events to a destination you set, and managed settings keep that destination fixed.

Rule Set telemetry destinations in managed settings, and keep prompt text off by default.

Trap A repository setting that enables telemetry looks simple. Claude Code ignores OTel variables in project settings, so the destination is not controlled.

Exam tip If the scenario asks for MCP call arguments, enable tool details and the logs exporter.

▼ RECALL CUES (3)

1. **a wrong action fired somewhere in a multi-step run and no one can find which step**
   Answer trace spans with one ID and cost per hop
   Rule Emit one trace per run, with a span for each model and tool hop.
   - More prompt logs
   - Longer log retention
2. **finance needs spend split by team and workspace**
   Answer usage and cost reports grouped by workspace
   Rule Call the usage and cost reports with group_by for workspace or key.
   - App-side estimates
   - Console screenshots for a monthly total
3. **developer tool use must be audited and the destination must stay fixed**
   Answer managed settings with OTel and tool detail events
   Rule Pin the telemetry destination in managed settings, and enable tool details for MCP audit.
   - Repository setting for telemetry
   - Prompt text logging for every session

▼ NUMBERS AND VERSIONS (5)

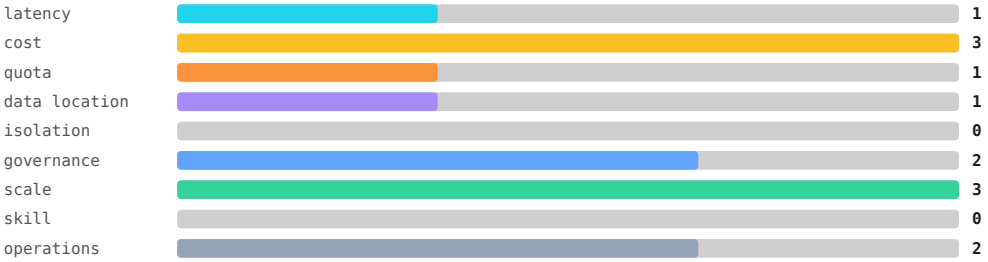
- Each span holds up to 0.75 MB of content by default. source
- The usage report default is 7 buckets for one-day width, and the maximum is 31. source
- A spend-cap 429 response has no retry-after header. source
- The default metric export interval is 60000 milliseconds. source
- The default logs export interval is 5000 milliseconds. source

D3.5 CHAPTER D3, INTEGRATION

The RAG pipeline

Search result blocks when answers need checkable sources.

CONSTRAINT FAMILIES



SEARCH RESULTS WITH CITATIONS

Pass retrieved text as search result blocks so Claude attaches a source and title to each cited claim.

Rule Return retrieved text as search result blocks when the answer needs verifiable source attribution.

Trap A prompt that asks the model to list sources in plain text looks simple. The names are not tied to retrieved blocks, so you cannot check them. (Inference)

Exam tip If the scenario requires cited claims, choose search result blocks with citations enabled.

EMBEDDINGS AND RERANKING MODELS

Convert text to vectors for similarity search, then reorder a short list with a reranker.

Rule Add a reranker when the answer appears in the candidates but ranks low. (Inference)

Trap An Anthropic embedding model looks like the default choice. Anthropic does not offer one, so the design needs an embeddings vendor.

Exam tip If the answer is absent from the candidates, fix retrieval before you add a reranker. (Inference)

CHUNK GRANULARITY FOR CITATIONS

Choose the document type whose chunk boundaries match the citation granularity you need.

Rule Use custom content when your own chunk boundaries must become the citation boundaries.

Trap Plain text documents look simpler, but automatic sentence chunking ignores table boundaries. (Inference)

Exam tip If the scenario names split tables, fix the chunker before you change citation settings. (Inference)

▼ RECALL CUES (3)

1. **`answer must name the source of each claim`**
   Answer search result blocks with citations enabled
   Rule Pass retrieved text as search results and enable citations.
   - A prompt that asks for sources in plain text
   - Citations combined with a strict JSON schema
2. **`answer cites the wrong section after a table`**
   Answer chunker fix at document structure
   Rule Fix the chunker first, then re-index. Do not swap the model.
   - Larger embedding model
   - Larger generation model
3. **`relevant passage is in the candidates but ranks low`**
   Answer reranking stage
   Rule Add a reranker that reorders the candidate list.
   - Larger generation model
   - Smaller chunk size without a rerank step

▼ NUMBERS AND VERSIONS (8)

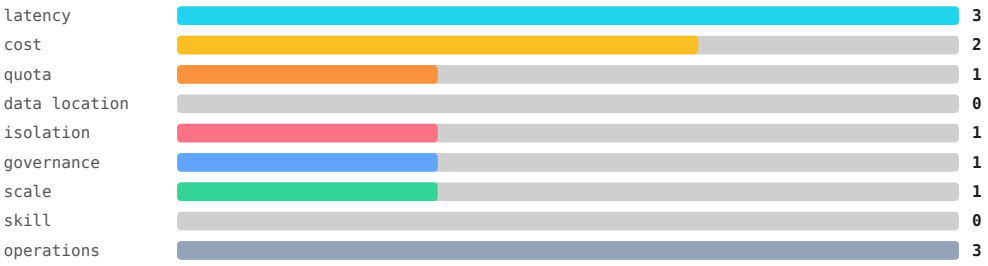
- All active models support search results with citations, except Claude Haiku 3. source
- Citations combined with structured outputs return a 400 error. source
- voyage-4 has a context length of 32,000 tokens and a default dimension of 1024. source
- voyage-context-4 accepts up to 120,000 tokens when enable_auto_chunking is true. source
- Voyage AI embeddings are normalized to length 1. source
- rerank-3 has a context length of 32,000 tokens and is recommended for most applications. source
- Int8 quantization reduces embedding storage by 4x compared with float. source
- Plain text citations use 0-indexed character indices with exclusive end indices. source

D3.6 CHAPTER D3, INTEGRATION

Retrieval strategies

Keywords for exact terms. Semantic for paraphrase. Filters for access.

CONSTRAINT FAMILIES



HYBRID KEYWORD AND SEMANTIC RETRIEVAL

Pick the matcher that fits the query shape, and merge the candidates from both matchers.

Rule Use keyword match for exact terms, semantic match for paraphrases, and filters for access. (Inference)

Trap A single semantic index looks sufficient. Exact codes often fail to match in vector space, so the codes go unfound. (Inference)

Exam tip If the scenario names both codes and paraphrases, choose hybrid retrieval. (Inference)

RERANKING CANDIDATES

A reranker scores a candidate list again and reorders it for the query.

Rule Add a reranker when the answer appears in the candidates but ranks low. (Inference)

Trap A larger generation model looks like the fix for an absent answer. The answer was never in the candidate list, so no model can cite it. (Inference)

Exam tip If the answer is absent from the candidates, fix retrieval, not the reranker. (Inference)

LIVE STATE THROUGH A TOOL CALL

Answer questions about current values with a tool call to the source system, not with an indexed snapshot.

Rule Route questions about live values to a tool, and route document questions to the index. (Inference)

Trap A faster index refresh looks like the fix for stale values. The index still stores a snapshot, so the answer can lag. (Inference)

Exam tip If the scenario names a value that changes in minutes, choose the tool path. (Inference)

▼ RECALL CUES (3)

1. **answer must reflect the current value of a record**
   Answer tool call to the source system
   Rule Read live state from a tool. Do not answer from a snapshot.
   - Faster index refresh
   - Larger model that reads the snapshot
2. **query contains an error code and also asks in plain words**
   Answer hybrid retrieval
   Rule Use keyword and semantic matchers together, then merge the results.
   - Semantic index only
   - Keyword index only
3. **relevant passage is retrieved but ranked low**
   Answer reranking stage
   Rule Add a reranker that reorders the candidate list.
   - Larger generation model
   - Wider index refresh

▼ NUMBERS AND VERSIONS (7)

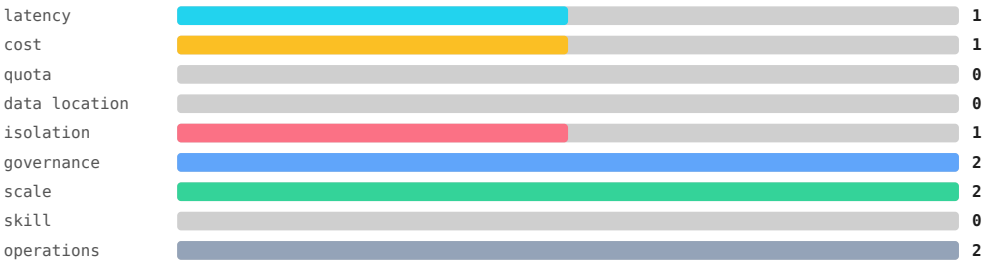
- Voyage AI embeddings are normalized to length 1. source
- Int8 embeddings cut storage by 4x compared with float. source
- Binary quantization cuts embedding storage by 32x compared with float. source
- rerank-3 has a context length of 32,000 tokens. source
- rerank-3-lite is optimized for latency and cost. source
- rerank-3 is the highest accuracy model in the listed rerankers. source
- The MCP connector uses the mcp-client-2025-11-20 beta header. source

D3.7 CHAPTER D3, INTEGRATION

Integration mechanisms

Messages API and remote MCP: connector, tool calls only.

CONSTRAINT FAMILIES



MCP CONNECTOR IN THE MESSAGES API

Connect a remote MCP server from the Messages API, so the API calls its tools without a client in your code.

Rule Use the MCP connector when a Messages API request needs tool calls on a remote MCP server.

Trap The connector looks like full MCP support. Only tool calls work, so resources and prompts need another path.

Exam tip If the scenario needs MCP resources or prompts, the connector alone does not cover it.

CLAUDE CODE MCP SERVERS

Register each MCP server once in Claude Code, with a transport that matches where the server runs.

Rule Use STDIO for a local process, and HTTP or SSE for a shared remote server.

Trap A local STDIO server looks like a fit for an API backend, but the Messages API connector cannot connect to it directly.

Exam tip If the scenario names a local command-line tool, choose a Claude Code STDIO server.

▼ RECALL CUES (3)

1. **one team and one host need one internal API**
   Answer direct API call
   Rule Skip the protocol when no second consumer exists.
   - MCP server for a single host
   - CLI wrapper with text parsing
2. **several hosts in several languages share one service**
   Answer MCP server with one schema
   Rule Build one server and let each host connect as a client.
   - Hand-rolled client per host
   - Slogan that MCP is the standard
3. **a Messages API request needs remote tools with no client code**
   Answer MCP connector
   Rule Define the server in mcp_servers, then reference it from one toolset.
   - Local STDIO server in the connector
   - Resources or prompts believed to work through the connector

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▼ NUMBERS AND VERSIONS (5)

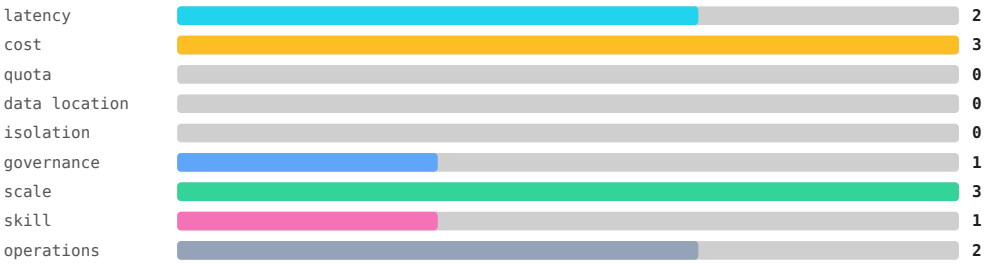
- The MCP connector uses the mcp-client-2025-11-20 beta header. source
- Batch processing gives a 50 percent discount on all usage. source
- A Message Batch holds up to 100,000 requests or 256 MB, whichever limit comes first. source
- From v2.1.265, Claude Code tries the HTTP transport first and switches to SSE when the server does not accept it. source
- The idle timeout defaults to five minutes for HTTP and SSE servers, and to 30 minutes for stdio servers. source

D3.8 CHAPTER D3, INTEGRATION

Progressive discovery and monolithic context

Ten or more tools, or over 10k tokens: defer the definitions.

CONSTRAINT FAMILIES



TOOL SEARCH TOOL IN THE MESSAGES API

Load only the tools that a request needs, found by a search tool over the full catalog.

Rule Defer tool definitions when the catalog reaches 10 or more tools, or when definitions exceed 10k tokens.

Trap Shorter tool descriptions look cheaper. The full schemas still load on each call, so the token cost stays high. (Inference)

Exam tip If the scenario counts tokens per call, choose deferral over description edits. (Inference)

AGENT SDK TOOL SEARCH

The Agent SDK withholds tool definitions by default and loads matches after the agent searches.

Rule Keep tool search on in the Agent SDK unless latency binds harder than tokens. (Inference)

Trap A fix that turns tool search off looks simple for wrong picks. Every definition then loads on every turn, so the token cost rises.

Exam tip If the scenario names a model that rejects tool search, load the tools upfront.

AGENT SKILLS PROGRESSIVE DISCLOSURE

Load skill metadata first, and read instructions and files only when a task needs them.

Rule Put only metadata in the prompt, and load each file when the task needs it.

Trap A system prompt that holds every reference file looks thorough. Each token then competes with the conversation for context space.

Exam tip If the scenario names large reference documents, choose skill files over a longer system prompt. (Inference)

▼ RECALL CUES (3)

1. **`agent costs 100,000 schema tokens per call with 200 tools`**
   Answer deferral with search
   Rule Defer definitions, keep a few frequent tools upfront, and search for the rest.
   - Rewrite all descriptions only
   - Remove tools from the config
2. **`wrong tool picks rise as the catalog grows`**
   Answer smaller visible choice set
   Rule Show fewer tools per turn, and load matches on demand.
   - Larger model that reads all schemas
   - Longer system prompt with every schema
3. **`long domain guidance must reach the agent only when relevant`**
   Answer skill with progressive disclosure
   Rule Keep metadata in the prompt, and read files on demand.
   - Full guidance in the system prompt
   - Script code pasted into context

▼ NUMBERS AND VERSIONS (13)

- Tool search returns up to 5 tools that match the query by default. source
- A request allows up to 10,000 tools with defer_loading true. source
- Regex patterns are limited to 200 characters, and BM25 queries to 500 characters. source
- The docs say tool selection accuracy degrades once you exceed 30 to 50 available tools. source
- The docs say tool search typically reduces tool definition tokens by over 85 percent. source
- The context window holds up to 1M tokens for several models, such as Claude Haiku 5.5. source
- The auto mode activates tool search when definitions reach 10 percent of the context window. source
- With alwaysLoad true, startup waits for the server tools, capped at the standard 5-second connect timeout. source
- Level 1 metadata costs about 100 tokens per skill. source
- Level 2 instructions cost under 5k tokens when a skill triggers. source
- The SKILL.md body should stay under 500 lines. source
- A skill name has a maximum of 64 characters. source
- A skill description has a maximum of 1,024 characters. source